Andreas Mang

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Full-Time Positions & Appointments

since 09/23 **Associate Professor of Mathematics**
Department of Mathematics, University of Houston, Houston, TX, US

09/17 – 08/23 **Assistant Professor of Mathematics**
Department of Mathematics, University of Houston, Houston, TX, US

05/13 – 08/17 **Postdoctoral Fellow**
Oden Institute for Computational Engineering & Sciences, University of Texas at Austin, Austin, TX, US

Other Positions & Activities

Co-Chair of SIAM MDS Conference
SIAM Conference on Mathematics of Data Science (MDS26), Salt Lake City, UT, US

ICERM Long Term Visitor (Spring 2026)
Institute for Computational and Experimental Research in Mathematics, Providence, RI, US

SIAM Coordinating Committee Member for JMM
Joint Mathematics Meetings 2026, Washington, DC, US

Co-Organizer of Local Conferences
SIAM TX-LA Sectional Meeting 2022
CMBS Conference: Research at the Interface of Applied Mathematics and Machine Learning 2025

Associate Editor
Advances in Continuous and Discrete Models (since 2023)
AIMS Journal of Numerical Algebra, Control and Optimization (since 2021)

Education

11/09 – 04/13 **Dr.-Ing. (Doctorate in Engineering Sciences)**
Division of Computer Science/Technology and Natural Sciences, University of Lübeck, Lübeck, DE

03/07 – 11/09 **MSc Biomedical Engineering**
University of Lübeck, Lübeck, DE

10/02 – 10/06 **Dipl.-Ing. Medical Engineering**
University of Applied Sciences Koblenz, RheinAhrCampus Remagen, Remagen, DE

Publications

Recent

[1] J. Y. Kim and **A. Mang**. Efficient Hessian and posterior covariance approximation for EPDiff-based diffeomorphic image registration. *Journal of Numerical Mathematics*, 2025. in preparation; 20 pages.

[2] A. Islam, R. B. Mizan, and **A. Mang**. Tensorial reduced-order models for multi-species brain tumor simulations. *International Journal for Numerical Methods in Biomedical Engineering*, 2025. in preparation; 30 pages.

[3] J. F. Chhoa, Y. He, and **A. Mang**. A multigrid-preconditioned Newton–Krylov method for diffeomorphic image registration. *SIAM Journal on Scientific Computing*, 2025. in preparation; 25 pages.

[4] J. Shi, N. Charon, **A. Mang**, R. Azencott, and D. Labate. Fast k-means clustering in Riemannian manifolds via Fréchet maps: Applications to large-dimensional SPD matrices. *SIAM Journal on Imaging Sciences*, 2025. in preparation; 35 pages.

[5] X. Wang, A. Valavanis, A. Mahmood, **A. Mang**, M. Benning, and A. Repetti. *A unified framework for lifted training and inversion approaches*. Machine Learning Solutions for Inverse Problems. 2025. [arXiv:2510.09796]; in review; 39 pages.

[6] Z. Zhou, B. Zhao, **A. Mang**, and G. Liao. VPreg: An optimal control formulation for diffeomorphic image registration based on the variational principle grid generation method. *SIAM Journal on Imaging Sciences*, 2025. [arXiv:2510.13109]; in review; 35 pages.

[7] G. Villalobos, J. Rudi, and **A. Mang**. Neural networks for Bayesian inverse problems governed by a nonlinear ODE. *SIAM Journal on Scientific Computing*, 2025. [arXiv:arXiv:2510.14197]; in review; 46 pages.

[8] Y. He and **A. Mang**. A generalized alternating NGMRES method for PDE-constrained optimization problems governed by transport equations. *SIAM Journal on Scientific Computing*, 2025. [arXiv:2510.08782]; in review; 22 pages.

Peer-Reviewed Journal Articles

[9] Y. Su, B. Geiger, I. Timofeyev, **A. Mang**, and R. Azencott. Rare events analysis and computation for stochastic evolution of bacterial populations. *Stochastic Analysis and Applications*, 41(1):1–29, 2025. [arXiv:2308.14135] [doi:10.1080/07362994.2024.2422913].

[10] H. Dabirian, R. Sultamuratov, J. Herring, C. El Tallawi, W. Zoghbi, **A. Mang**, and R. Azencott. Classification of deformable smooth shapes through geodesic flows of diffeomorphisms. *Journal of Mathematical Imaging and Vision*, 66:1033–1059, 2024. [arXiv:2211.02530] [doi:10.1007/s10851-024-01211-z].

[11] **A. Mang**, J. He, and R. Azencott. An operator-splitting approach for variational optimal control formulations for diffeomorphic shape matching. *Journal of Computational Physics*, 493(112463), 2023. [arXiv:2307.10114] [doi:10.1016/j.jcp.2023.112463].

[12] N. Himthani, M. Brunn, J. Y. Kim, M. Schulte, **A. Mang**, and G. Biros. CLAIRE: Parallelized diffeomorphic image registration for large-scale biomedical imaging applications. *Journal of Imaging*, 8(9):251, 2022. [arXiv:2209.08189] [doi:10.3390/jimaging8090251].

[13] S. Sarmadi, J. J. Winkle, R. N. Alnahhas, M. R. Bennett, K. Josić, **A. Mang**, and R. Azencott. Stochastic neural networks for automatic cell tracking in microscopy image sequences of bacterial colonies. *Mathematical and Computational Applications*, 27(2):22, 2022. [arXiv:2104.13482] [doi:10.3390/mca27020022].

[14] M. Brunn, N. Himthani, G. Biros, M. Mehl, and **A. Mang**. CLAIRE: Constrained large deformation diffeomorphic image registration on parallel computing architectures. *Journal of Open Source Software*, 6(61):3038, 2021. [doi:10.21105/joss.03038].

[15] **A. Mang**, S. Bakas, S. Subramanian, C. Davatzikos, and G. Biros. Integrated biophysical modeling and image analysis: Application to neuro-oncology. *Annual Review of Biomedical Engineering*, 22, 2020. [arXiv:2002.09628] [doi:10.1146/annurev-bioeng-062117-121105].

[16] P. Zhang, **A. Mang**, J. He, R. Azencott, K. C. El-Tallawi, and W. A. Zoghbi. Diffeomorphic shape matching by operator splitting in 3D cardiology imaging. *Journal of Optimization Theory and Applications*, 188:143–168, 2021. [arXiv:2011.10165] [doi:10.1007/s10957-020-01789-5].

[17] M. Brunn, N. Himthani, G. Biros, M. Mehl, and **A. Mang**. Fast GPU 3D diffeomorphic image registration. *Journal of Parallel and Distributed Computing*, 149:149–162, 2021. [arXiv:2004.08893] [doi:10.1016/j.jpdc.2020.11.006].

[18] K. Scheufele, S. Subramanian, **A. Mang**, G. Biros, and M. Mehl. Image-driven biophysical tumor growth model calibration. *SIAM Journal on Scientific Computing*, 42(3):B549–B580, 2020. [arXiv:1907.07774] [doi:10.1137/19M1275280].

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[20] **A. Mang**, A. Gholami, C. Davatzikos, and G. Biros. CLAIRE: A distributed-memory solver for constrained large deformation diffeomorphic image registration. *SIAM Journal on Scientific Computing*, 41(5):C548–C584, 2019. [arXiv:1808.04487] [doi:10.1137/18M1207818].

[21] **A. Mang**, A. Gholami, C. Davatzikos, and G. Biros. PDE-constrained optimization in medical image analysis. *Optimization and Engineering*, 19(3):765–812, 2018. [arXiv:1803.00058] [doi:10.1007/s11081-018-9390-9].

[22] **A. Mang** and L. Ruthotto. A Lagrangian Gauss–Newton–Krylov solver for mass- and intensity-preserving diffeomorphic image registration. *SIAM Journal on Scientific Computing*, 39(5):B860–B885, 2017. [arXiv:1703.04446], [doi:10.1137/17M1114132].

[23] **A. Mang** and G. Biros. A semi-Lagrangian two-level preconditioned Newton–Krylov solver for constrained diffeomorphic image registration. *SIAM Journal on Scientific Computing*, 39(6):B1064–B1101, 2017. [arXiv:1604.02153] [doi:10.1137/16M1070475].

[24] A. Gholami, **A. Mang**, and G. Biros. An inverse problem formulation for parameter estimation of a reaction-diffusion model for low grade gliomas. *Journal of Mathematical Biology*, 72(1):409–433, 2016. [arXiv:1408.6221], [doi:10.1007/s00285-015-0888-x].

[25] **A. Mang** and G. Biros. Constrained H^1-regularization schemes for diffeomorphic image registration. *SIAM Journal on Imaging Sciences*, 9(3):1154–1194, 2016. [arXiv:1503.00757], [doi:10.1137/15M1010919].

[26] **A. Mang** and G. Biros. An inexact Newton–Krylov algorithm for constrained diffeomorphic image registration. *SIAM Journal on Imaging Sciences*, 8(2):1030–1069, 2015. [arXiv:1408.6299], [doi:10.1137/140984002].

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[30] A. Toma, **A. Mang**, T. A. Schuetz, S. Becker, and T. M. Buzug. A novel method for simulating the extracellular matrix in models of tumour growth. *Computational and Mathematical Methods in Medicine*, 2012:960256–1–960256–11, 2012. [doi:10.1155/2012/109019].

[31] **A. Mang**, A. Toma, T. A. Schuetz, S. Becker, C. Mohr, T. Eckey, D. Petersen, and T. M. Buzug. Biophysical modeling of brain tumor progression: From unconditionally stable explicit time integration to an inverse problem with parabolic PDE constraints for model calibration. *Medical Physics*, 39(7):4444–4460, 2012. [doi:10.1118/1.4722749].

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[33] S. Becker, **A. Mang**, A. Toma, and T. M. Buzug. In-silico oncology: An approximate model of brain tumor mass effect based on directly manipulated free form deformation. *International Journal of Computer Assisted Radiology and Surgery*, 5(6):607–622, 2010. [doi:10.1007/s11548-010-0531-7].

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Peer-Reviewed Conference Papers

[36] M. Brunn, N. Himthani, G. Biros, M. Mehl, and **A. Mang**. Multi-node multi-GPU diffeomorphic image registration for large-scale imaging problems. In *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*, pages 523–539, 2020. [arXiv:2008.12820] [doi:10.1109/SC41405.2020.00042].

[37] A. Gholami, **A. Mang**, K. Scheufele, C. Davatzikos, M. Mehl, and G. Biros. A framework for scalable biophysics-based image analysis. In *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*, number 19, pages 1–13, 2017. [doi:10.1145/3126908.3126930] (**Best Student Paper Award**; Laureate: A. Gholami).

[38] **A. Mang**, A. Gholami, and G. Biros. Distributed-memory large-deformation diffeomorphic 3D image registration. In *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*, number 72, pages 842–853, 2016. [arXiv:1608.03630], [web].

[39] T. A. Schuetz, S. Becker, **A. Mang**, A. Toma, and T. M . Buzug. A computational multiscale model of glioblastoma growth: Regulation of cell migration and proliferation via microRNA-451, LKB1 and AMPK. In *Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, pages 6620–6623, 2012. [doi:10.1109/EMBC.2012.6347512].

Other Conference & Workshop Papers

[40] J. Y. Kim and **A. Mang**. Efficient numerical schemes for uncertainty quantification in diffeomorphic image registration governed by transport equations. In *International Conference on Computational and Mathematical Biomedical Engineering*, volume 2, pages 326–329, 2024.

[41] J. Rudi, G. Villalobos, and **A. Mang**. Data representation for parameter estimation with deep learning models for a dynamical system. In *International Conference on Computational and Mathematical Biomedical Engineering*, volume 1, pages 157–160, 2024.

[42] S. Pati, V. Sharma, H. Aslam, S. Thakur, H. Akbari, **A. Mang**, S. Subramanian, G. Biros, and C. Davatzikos. Estimating glioblastoma biophysical growth parameters using deep learning regression. In *Brainlesion: Glioma, Multiple Sclerosis, Stroke and Traumatic Brain Injuries. BrainLes 2020*, volume LNCS 12658, pages 157–167, 2021. [doi:10.1007/978-3-030-72084-1_15].

[43] M. Brunn, N. Himthani, G. Biros, M. Mehl, and **A. Mang**. Fast 3D diffeomorphic image registration on GPUs. In *The International Conference for High Performance Computing, Networking, Storage, and Analysis*, pages 1–3, 2019. [web].

[44] N. Himthani, **A. Mang**, A. Gholami, and G. Biros. GPU-accelerated interpolation for 3D image registration. In *The International Conference for High Performance Computing, Networking, Storage, and Analysis*, pages 1–3, 2018. [web].

[45] **A. Mang**, S. Tharakan, A. Gholami, N. Nimthani, S. Subramanian, J. Levitt, M. Azmat, K. Scheufele, M. Mehl, C. Davatzikos, B. Barth, and G. Biros. SIBIA-GIS: Scalable biophysics-based image analysis for glioma segmentation. In *Proceedings of the BraTS 2017 Workshop (MICCAI)*, pages 197–204, 2017.

[46] **A. Mang**, J. Stritzel, A. Toma, S. Becker, T. A. Schuetz, and T. M. Buzug. Personalisierte Modellierung der Progression primärer Hirntumoren als Optimierungsproblem mit Differentialgleichungsnebenbedingungen. In *Bildverarbeitung für die Medizin*, pages 57–62, 2013. [doi:10.1007/978-3-642-36480-8_12].

[47] A. Toma, K. Holl-Ulrich, S. Becker, **A. Mang**, T. A. Schuetz, M. M. Bonsanto, V. Tronnier, and T. M. Buzug. A mathematical model to simulate glioma growth and radiotherapy at the microscopic level. In *Biomedizinische Technik*, pages 218–221, 2012. [doi:10.1515/bmt-2012-4081].

[48] **A. Mang**, A. Toma, T. A. Schütz, S. Becker, and T. M. Buzug. Eine effiziente Parallel-Implementierung eines stabilen Euler-Cauchy-Verfahrens fuer die Modellierung von Tumorwachstum. In *Bildverarbeitung für die Medizin*, pages 63–68, 2012. [doi:10.1007/978-3-642-28502-8_13].

[49] **A. Mang**, T. A. Schütz, A. Toma, S. Becker, and T. M. Buzug. Ein dämonenartiger Ansatz zur Modellierung tumorinduzierter Gewebedeformation als Prior für die nicht-rigide Bildregistrierung. In *Bildverarbeitung für die Medizin*, pages 422–427, 2012. [doi:10.1007/978-3-642-28502-8_73].

[50] T. A. Schütz, S. Moeller, S. Becker, **A. Mang**, and A. Toma. A cross-scale model of tumor growth: Do we need to model molecular interactions in separate artificial compartments within a cell? In *IFAC Proceedings Volumes*, pages 1294–1299, 2012. [doi:10.3182/20120215-3-AT-3016.00230] (**Best Poster Award**; Category Students; Laureate: A. Schütz).

[51] A. Toma, A. Régnier-Vigouroux, **A. Mang**, S. Becker, T. A. Schütz, and T. M. Buzug. In-silico modelling of tumour-immune system interactions for glioblastomas. In *IFAC Proceedings Volumes*, volume 45, pages 1237–1242, 2012. [doi:10.3182/20120215-3-AT-3016.00219] (**Best Poster Award**; Category Work in Progress; Laureate: A. Toma).

[52] **A. Mang**, T. A. Schütz, A. Toma, S. Becker, and T. M. Buzug. An efficient, variational non-parametric model of tumour induced brain deformation to aid non-diffeomorphic image registration. In *IEEE International Symposium on Biomedical Imaging: From Nano to Macro*, pages 732–735, 2012. [doi:10.1109/ISBI.2012.6235652].

[53] **A. Mang**, T. A. Schuetz, S. Becker, A. Toma, and T. M. Buzug. Cyclic numerical time integration in variational non-rigid image registration based on quadratic regularisation. In *Proceedings of the Vision Modeling and Visualization Workshop*, pages 143–150, 2012. [doi:10.2312/PE/VMV/VMV12/143-150].

[54] A. Toma, A. Régnier-Vigouroux, **A. Mang**, T. A. Schütz, S. Becker, and T. M. Buzug. In-silico Modellierung der Immunantwort auf Hirntumorwachstum. In *Bildverarbeitung für die Medizin*, pages 123–128, 2012. [doi:10.1007/978-3-642-28502-8_23] (**Best Poster Award**; Laureate: A. Toma).

[55] Y. Schröder, S. Becker, A. Toma, **A. Mang**, A. Schuetz, and T. M. Buzug. Ein diskreter Ansatz zur Modellierung von Tumorwachstum und Strahlentherapie. In *Bildverarbeitung für die Medizin*, pages 379–383, 2011. [doi:10.1007/978-3-642-19335-4_78].

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[57] **A. Mang**, S. Becker, A. Toma, T. A. Schütz, J. Küchler, V. Tronner, M. M. Bonsanto, and T. M. Buzug. A model of tumour induced brain deformation as bio-physical prior for non-rigid image registration. In *IEEE International Symposium on Biomedical Imaging: From Nano to Macro*, pages 578–581, 2011. [doi:10.1109/ISBI.2011.5872473].

[58] A. Toma, **A. Mang**, T. A. Schütz, S. Becker, and T. M. Buzug. Is it necessary to model the matrix degrading enzymes for simulating tumour growth? In *Proceedings of the Vision Modeling and Visualization Workshop*, pages 361–368, 2011. [doi:10.2312/PE/VMV/VMV11/361-368].

[59] **A. Mang**, S. Becker, A. Toma, T. A. Schütz, and T. M. Buzug. Modellierung tumorinduzierter Gewebedeformation als Optimierungsproblem mit weicher Nebenbedingung. In *Bildverarbeitung für die Medizin*, pages 294–298, 2011. [doi:10.1007/978-3-642-19335-4_61] (**Best Scientific Contribution**; Ranked #3).

[60] S. Becker, **A. Mang**, and T. M. Buzug. Approximation des Tumormasseeffekts mittels direkt-manipulierender Free-Form Deformation. In *Bildverarbeitung für die Medizin*, pages 306–310, 2010.

[61] **A. Mang**, S. Becker, A. Toma, and T. M. Buzug. Coupling tumor growth with brain deformation: a constrained parametric non-rigid registration problem. In *Proceedings of SPIE Medical Imaging: Image*

Processing, volume 7623, pages 76230C 1–12, 2010. [doi:10.1117/12.844107] (**Best Paper Award**; FIT).

[62] S. Becker, **A. Mang**, A. Toma, and T. M. Buzug. Approximating tumor induced brain deformation using directly manipulated free form deformation. In *IEEE International Symposium on Biomedical Imaging: From Nano to Macro*, pages 85–88, 2010. [doi:10.1109/ISBI.2010.5490409].

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[64] S. Becker, **A. Mang**, J. O. Jungmann, and T. M. Buzug. In-silico Modellierung von Tumorwachstum: Approximation des Tumormasseeffektes mittels Thin-Plate-Splines. In *Informatik 2009–Im Focus das Leben*, pages 1276–1284, 2009.

[65] S. Becker, J. O. Jungmann, **A. Mang**, and T. M. Buzug. Tumor-Wachstumsmodellierung als parametrisches Bildregistrierproblem. In *Bildverarbeitung für die Medizin*, pages 197–201, 2009. [doi:10.1007/978-3-540-93860-6_40].

[66] **A. Mang**, J. Müller, and T. M. Buzug. Soft-tissue segmentation in forensic applications. In *Facial Reconstruction (Police and Research 35)*, pages 62–77, 2007.

[67] J. Müller, **A. Mang**, and T. M. Buzug. Radial basis functions for 3D nonlinear soft-tissue warping. In *Facial Reconstruction (Police and Research 35)*, pages 161–175, 2007.

[68] **A. Mang**, O. Camara, G. B. Caseiras, W. R. Crum, J. A. Schnabel, T. M. Buzug, J. Thornton, H. R. Jäger, and D. J. Hawkes. Registration of rCBV and ADC maps with structural and physiological MR images in glioma patients: study and validation. In *IEEE International Symposium on Biomedical Imaging: From Nano to Macro*, pages 37–40, 2007. [doi:10.1109/ISBI.2007.356782].

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[70] **A. Mang**, W. R. Crum, O. Camara-Rey, J. A. Schnabel, G. P. Penney, G. B. Caseiras, T. M. Buzug, H. R. Jäger, T. A. Yousry, and D. J. Hawkes. Modelling tumour growth patterns with non-rigid image registration. In *Springer Proceedings in Physics 114*, pages 139–144, 2007. [doi:10.1007/978-3-540-68764-1_23].

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[72] **A. Mang**, J. Müller, and T. M. Buzug. Gradient vector flow based active contours for facial reconstruction. In *Aktuelle Methoden der Laser- und Medizinphysik*, pages 218–223, 2005.

[73] J. Müller, **A. Mang**, and T. M. Buzug. 3D warping for forensic soft-tissue reconstruction. In *Aktuelle Methoden der Laser- und Medizinphysik*, pages 212–217, 2005.

Books & Book Chapters

[74] **A. Mang**. *CLAIRE: Scalable GPU-Accelerated Algorithms for Diffeomorphic Image Registration in 3D*, pages 167–215. Applied and Numerical Harmonic Analysis. Springer, 2024. ([arXiv:2401.17493] [doi:10.1007/978-3-031-66497-7_8]).

[75] **A. Mang**. *Methoden zur numerischen Simulation der Progression von Gliomen: Modellentwicklung, Numerik und Parameteridentifikation*. Springer Verlag, 2014. [doi:10.1007/978-3-658-05246-1].

Theses & Reports

[76] G. Biros, **A. Mang**, B. H. Menze, and M. Schulte. Inverse Biophysical Modeling and Machine Learning in Personalized Oncology (Dagstuhl Seminar 23022). *Dagstuhl Reports*, 13(1):36–67, 2023. [doi:10.4230/DagRep.13.1.36].

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[79] **A. Mang**. Parametric deformation models in image computing. Master's thesis, Division of Computer Science/Technology and Natural Sciences, University of Lübeck, 2009.

[80] **A. Mang**. Medical image registration to improve the understanding of disease progression of glioma tumours. Diploma thesis, Department of Mathematics and Technolgy, RheinAhrCampus Remagen & Department of Computer Science, Centre for Medical Image Computing, University College London, 2006.

[81] **A. Mang**. Inflating the brain. Technical report, Department for Mathematics and Technology, RheinAhrCampus Remagen, Remagen, DE, 2005.

Grants, Honors, & Awards

Awarded Extramural Research Grants

- **CAREER: Scalable algorithms for nonlinear, large-scale inverse problems governed by dynamical systems (DMS 2145845).**
  Funding Agency: National Science Foundation (NSF; Division of Mathematical Sciences) • *Program:* Computational Mathematics (program solicitation NSF 20-525) • *CFDA:* 47.049 Mathematical and Physical Sciences • *PI:* A. Mang (UHouston; Single Investigator) • *Awarded Funds:* ∼$100K / year for 5 years (total: ∼$500K) • *Funding Period:* 08/01/2022–07/31/2027.
- **Analysis and simulation of extremes and rare events in complex systems (DMS 2009923).**
  Funding Agency: National Science Foundation (NSF; Division of Mathematical Sciences) • *Program:* Applied Mathematics (program solicitation PD 16-1266) • *CFDA:* 47.049 Mathematical and Physical Sciences • *PI:* M. Nicol (UHouston) • *Co-PIs:* R. Azencott (UHouston), A. Mang (UHouston) • *Awarded Funds:* ∼$120K / year for 3 years (total: ∼$370K) • *Funding Period:* 07/15/2020–06/30/2023.
- **Fast algorithms for nonlinear optimal control of geodesic flows of diffeomorphisms (DMS 2012825).**
  Funding Agency: National Science Foundation (NSF; Division of Mathematical Sciences) • *Program:* Computational Mathematics (program solicitation PD 16-127) • *CFDA:* 47.049 Mathematical and Physical Sciences • *PI:* A. Mang (UHouston; Single Investigator) • *Awarded Funds:* ∼$95K / year for 3 years (total: ∼300K) • *Funding Period:* 07/01/2020–06/30/2023.
- **Data mining for large data sets of shape deformations (DMS 1854853).**
  Funding Agency: National Science Foundation (NSF; Division of Mathematical Sciences) • *Program:* Computational and Data-Enabled Science and Engineering in Mathematical and Statistical Sciences (CDS&E-MSS; program solicitation PD 16-8069) • *CFDA:* 47.049 Mathematical and Physical Sciences • *PI:* R. Azencott (UHouston) • *Co-PIs:* J. He (UHouston), A. Mang (UHouston) • *Awarded Funds:* ∼$135K / year for 3 years (total: ∼$400K) • *Funding Period:* 08/01/2019–07/31/2022.
- **Integrated methods for data analytics and inversion in image computing.**
  Funding Agency: Simons Foundation • *Program:* Mathematics and Physical Sciences-Collaboration Grants for Mathematicians • *PI:* A. Mang (UHouston; Single Investigator) • *Awarded Funds:* ∼$8K / year for 5 years (total: $42K) • *Funding Period:* 09/01/2018–08/31/2019 (award expired due to NSF DMS-1854853).

Awarded Internal Research Grants

- **Fast clustering of medical neuroimaging data on Riemannian manifolds.**
  Sponsor: Division of Research, University of Houston • *Program:* Grants to Enhance and Advance Research • *PI:* D. Labate (UHouston) • *Co-PI:* A. Mang (UHouston) • *Requested Funds:* ∼$40K for 8 months. *Funding Period:* 05/15/2025–11/15/2025.

Awarded Conference & Workshop Grants

- **CBMS Conference: Research at the interface of applied mathematics and machine learning (DMS 2430460).**
  Funding Agency: National Science Foundation (NSF; Division of Mathematical Sciences) • *Program:* NSF-CBMS Regional Research Conferences in the Mathematical Sciences (program solicitation NSF 19-539) • *Organizers:* Y. He (UHouston), L. Cappanera (UHouston), A. Mang (UHouston), M. Wang (UHouston) • *Awarded Funds:* $42K • *Venue:* University of Houston, Houston, TX, US • *Funding Period:* 06/01/2025–05/31/2026.

Awarded Workshop Proposals

- **Integrating data- and physics-driven methods for decision making under uncertainty (26w5632).**
  Type: BIRS Workshop (26w5632) • *Sponsor:* BIRS • *Organizers:* T. Bui-Thanh (UTAustin); A. Mang (UHouston); L. Leticia Ramirez Ramirez (CIMAT); R. White (Sandia) • *Venue:* CMO • *Dates:* 05/31/2025–06/05/2026.
- **Inverse biophysical modeling and machine learning (23022).**
  Type: Dagstuhl Seminar • *Sponsor:* Leibniz Center for Informatics • *Organizers:* George Biros (UTAustin); Andreas Mang (UHouston); Miriam Schulte (UStuttgart); Bjoern Menze (UZuerich) • *Venue:* Schloss Dagstuhl, Leibniz-Center for Informatics, Wadern, DE • *Dates:* 01/08–01/13, 2023.

Awarded Hardware & Allocations

- **Cerebras accelerated deep neural networks for parameter estimation in scientific models.**
  Program: Neocortex Spring 2023 Call • *Team:* J. Rudi (VTech), A. Mang (UHouston), G. Villalobos (UHouston) • *Award:* Allocation on the Neocortex Supercomputer at the Pittsburgh Supercomputing Center (to access the integrated technologies of the Cerebras CS-2 and the HPE Superdome Flex Servers) • *Time Period:* 06/01/2023–05/31/2024.
- **GPU-accelerated solvers for nonlinear optimal control problems**
  Sponsor: NVIDIA Corporation • *Program:* GPU Grant Program (Accelerated Data Science Call 2019) • *PI:* A. Mang (UHouston; Single Investigator) • *Awarded Hardware:* 2 NVIDIA Titan V VOLTA 12GB HBM2 Video Cards (∼$6K).

Awards & Honors

- **Best Poster Award** **2025**
  Poster Title: Efficient Numerical Methods for Multi-Species Tumor Growth Simulations • *Authors:* A. Islam & A. Mang • *Venue:* SIAM TX-LA Regional Meeting • *Role:* Adviser.
- **Award for Excellence in Research, Scholarship and Creative Activity** **2023**
  Division of Research, University of Houston
- **NSM Junior Faculty Award for Excellence in Research** **2022**
  College of Natural Sciences and Mathematics, University of Houston
- **NSF CAREER Award** **2022**
  National Science Foundation (NSF), Division of Mathematical Sciences, Computational Mathematics Program

- **Best Student Paper Award** **2017**
  Publication: A framework for scalable biophysics-based image analysis • *Authors:* A. Gholami, A. Mang, K. Scheufele, C. Davatzikos, M. Mehl & G. Biros • *Venue:* International Conference for High Performance Computing, Networking, Storage and Analysis, Denver, CO, USA • *Role:* Co-Author.
- **Best Poster Award** **2012**
  Publication: In-silico modelling of tumour-immune system interactions for glioblastomas • *Authors:* A. Toma, A. Régnier-Vigouroux, A. Mang, S. Becker, T. A. Schütz, & T. M. Buzug • *Venue:* MathMod, Vienna, AT • *Role:* Co-Author.
- **Best Poster Award** **2012**
  Publication: A cross-scale model of tumour growth: Do we need to model molecular interactions in separate artificial compartments within a cell? • *Authors:* T. A. Schütz, S. Moeller, S. Becker, A. Mang & A. Toma • *Venue:* MathMod, Vienna, AT • *Role:* Co-Author.
- **Best Student Contribution** **2012**
  Publication: A mathematical model to simulate glioma growth and radiotherapy at the microscopic level • *Authors:* A. Toma, K. Holl-Ulrich, S. Becker, A. Mang, T. A. Schütz, M. M. Bonsanto, V. Tronnier & T. M. Buzug • *Venue:* Biomedizinische Technik, Jena, DE • *Role:* Co-Author.
- **Best Poster Presentation**
  Publication: In-silico Modellierung der Immunantwort auf Hirntumorwachstum • *Authors:* A. Toma, A. Régnier-Vigouroux, A. Mang, S. Becker, T. A. Schütz & T. M. Buzug • *Venue:* Bildverarbeitung für die Medizin, Berlin, DE • *Role:* Co-Author.
- **Best Scientific Contribution** **2011**
  Publication: Modellierung tumorinduzierter Gewebedeformation als Optimierungsproblem mit weicher Nebenbedingung • *Authors:* A. Mang, S. Becker, A. Toma, T. Polzin, T. A. Schütz & T. M. Buzug • *Venue:* Bildverarbeitung für die Medizin, Lübeck, DE • *Role:* Co-Author.
- **Best Student Contribution** **2011**
  Publication: Ein kontinuierlicher Ansatz zur Modellierung von Tumorwachstum und Strahlentherapie • *Authors:* A. Heye, S. Becker, A. Mang, T. A. Schütz, A. Toma & T. M. Buzug • *Venue:* Biomedizinische Technik, Freiburg, DE • *Role:* Co-Author.
- **Best Scientific Publication** **2010**
  Publication: Coupling tumor growth with brain deformation:A constrained parametric non-rigid registration problem • *Authors:* A. Mang, S. Becker, A. Toma & T. M. Buzug • *Venue:* Forum Innovative Therapies, Hamburg, DE • *Role:* Co-Author.
- **Best Supraregional Diploma Thesis** **2006**
  Publication: Medical image registration to improve the understanding of disease progression of glioma tumours • *Sponsor:* Society for the Promotion of RheinAhrCampus Remagen, Remagen, DE.

Financial Support & Workshop Invitations

2026 **Long Term Visitor at ICERM's semester program "Stochastic and Randomized Algorithms in Scientific Computing: Foundations and Applications."**
Organizers: H. Antil (George Mason), J. Chung (Emory), P. Drineas (Purdue); Y. Marzouk (MIT); A. Miedlar (Virginia Tech) & A. K. Saibaba (NC State) • *Venue:* Institute for Computational and Experimental Research in Mathematics (ICERM), Brown University, Providence, RI, US • *Sponsor:* ICERM.

2025 **Selected and financially supported to participate in the IMSI Workshop "Optimal Control and Decision Making Under Uncertainty for Digital Twins."**
Organizers: G. Stadler (NYU), M. Diehl (UFreiburg), M. Heinkenschloss (Rice), X. Huan (UMichigan) & C. Schillings (FUB) • *Venue:* The Institute for Mathematical and Statistical Innovation (IMSI), Chicago, IL, US • *Sponsor:* IMSI.

2023 **Invitation and financial support to participate in INdAM Workshop "Learning for Inverse Problems."**
Organizers: G. S. Alberti (UGenoa), A. Aspri (UMilan), T. Bubba (UBath), L. Ratti (UGenova) & M. Santacesaria (UGenoa) • *Venue:* Istituto Nazionale di Alta Matematica Francesco Severi, Rome, IT • *Sponsor:* INdAM.

Invitation and financial support to participate in BIRS Workshop "Leveraging Model- and Data-Driven Methods in Medical Imaging."
Organizers: T. Bubba (UBath), Daniel Vera (ITAM) & D. Labate (UHouston) • *Venue:* Banff International Research Station for Mathematical Innovation and Discovery, Kelowna, British Columbia, CA • *Sponsor:* BIRS.

2020 **Selected and financially supported to participate in IPAM Workshop "PDE and Inverse Problem Methods in Machine Learning."**
Organizers: A. Oberman (McGill), L. Rosasco (UGenoa), D. Slepcev (Carnegie Mellon), A. Stuart (CalTech) & Y. Yang (NYU) • *Venue:* Institute for Pure and Applied Mathematics (IPAM), University of California, Los Angeles, CA, US • *Sponsor:* IPAM.

2019 **Selected and financially supported to participate in ICERM Workshop "Optimization Methods in Computer Vision and Image Processing."**
Organizers: Y. Boykov (UWaterloo), P. Felzenszwalb (Brown), B. Fux (IMPA) & O. Veksler (UWaterloo) • *Venue:* Institute for Computational and Experimental Research in Mathematics (ICERM), Brown University, Providence, RI, US • *Sponsor:* ICERM.

Invitation and financial support to participate to Workshop "New Trends in PDE-Constrained Optimization" at RICAM's Special Semester on Optimization.
Organizers: R. Herzog (TU Chemnitz) & E. Trélat (USorbonne) • *Organizers (Special Semester):* K. Kunisch (UGraz & RICAM) & E. Sachs (UTrier) • *Venue:* Johann Radon Institute for Computational and Applied Mathematics (RICAM), Linz, Austria • *Sponsor:* RICAM.

2015 **Selected as participant of the IdeaLab program "Inverse Problems and Uncertainty Quantification" (15 participants).**
Organizers: O. Ghattas (UTAustin), Y. Marzouk (MIT) and N. Petra (UCMerced) • *Venue:* Institute for Computational and Experimental Research in Mathematics (ICERM), Brown University, Providence, RI, US • *Sponsor:* ICERM.

2010 **Selected as participant of the workshop/summer school on "Mathematical Modeling of Cancer Growth and Treatment" (56 participants)**
Venue: University of Dundee, Dundee, UK • *Sponsor:* University of Dundee, Dundee, UK.

2006 **Foreign exchange scholarship (diploma thesis)**
Venue: Centre for Medical Image Computing, University College London, London, UK • *Sponsor:* German Academic Exchange Service (DAAD).

Invited & Contributed Talks

"Invited Talks" are talks that were financially supported by the host or hosting institution. "Contributed Talks" are any talks that were not financially supported, invited (marked with ∗) or contributed.

84. **Fast algorithms for inverse transport problems**.
Invited talk (host: I. Tomas) at Department of Mathematics & Statistics, Texas Tech University, Lubbock, TX, US, 2025.

83. **A generalized alternating nonlinear GMRES acceleration method**.
Contributed* talk at SIAM Texas-Louisiana Sectional Meetings (SIAM TX-LA;; Session: *High-Performance Solvers and Rapid PDE-Constrained Optimization*), University of Texas at Austin, Austin, TX, US, 2025

82. **Numerical methods for PDE-based diffeomorphic image registration**.
Contributed* talk at SIAM Annual Meeting (SIAMAN25; Session: *Image Analysis and Learning with Variational Models and PDEs*), Montreal, CA, 2025.

81. **CLAIRE: Constrained large deformation diffeomorphic image registration**.
Contributed* talk at International Conference on Continuous Optimization (ICCOPT; Session: *Recent Advances on PDE-constrained optimization packages and libraries*), University of Southern California, Los Angeles, CA, US, 2025.

80. **Data- and model-driven approaches for solving inverse problems**.
Invited talk (hosts: D. Mishra (TAMU), M. Zhong (UH), X. Chen (TAMU), D. Casey (TAMU)) at the *Scientific Machine Learning (SciML) Summer School* at the Institute of Data Science, Texas AM, College Station, TX, US, 2025.

79. **Bayesian inference for large scale inverse problems governed by hyperbolic dynamical systems**.
Contributed* talk at SIAM Conference on Computational Sciences and Engineering (CSE 25; Session: *Investigating Inverse Problems using Bayesian Inference: Challenges and Advances*); Fort Worth, TX, US, 2025.

78. **Efficient numerical methods for inverse problems governed by transport equations**.
Contributed* talk at 3rd IACM Digital Twins in Engineering Conference (DTE 2025) & 1st ECCOMAS Artificial Intelligence and Computational Methods in Applied Science (DTE & AICOMAS 25; Session: *Inverse Problems and Data Assimilation for Digital Twins*); Paris, FR, 2025.

77. **Deep learning for Bayesian inverse problems governed by nonlinear ODEs**.
Contributed* talk at SIAM Conference on Mathematics of Data Science (MDS24; Session: *Recent Advances in Scientific Deep Learning*); Atlanta, GA, US, 2024.

76. **Fast iterative methods for large-scale initial value control problems**.
Contributed* talk at SIAM Texas-Louisiana Sectional Meetings (SIAM TX-LA;; Session: *Recent Developments in Computational Inversion and Reduced Order Modelling*), Baylor University, Waco, TX, US, 2024.

75. **Fast iterative methods for large-scale initial value control problems**.
Contributed* talk at the Modeling and Optimization: Theory and Applications Conference (MOPTA; Session: *Computational and Theoretical Methods for High-dimensional Optimization Problems*), Lehigh University, Bethlehem, PA, USA, 2024.

74. **Efficient numerical schemes for uncertainty quantification in diffeomorphic image registration governed by transport equations**.
Contributed talk at the International Conference on Computational and Mathematical Biomedical Engineering (CMBE24; Session: *Inverse Problems and Uncertainty Quantification in Biological and Medical Applications*), Arlington, VA, USA, 2024.

73. **CLAIRE: Scalable algorithms for diffeomorphic image registration**.
Contributed talk at the SIAM Conference on Imaging Sciences (IS24; Session: *Model- and Data-Driven Approaches in Motion Analysis*), Atlanta, US, 2024.

72. **Fast iterative solvers for initial value control problems with application to diffeomorphic image registration**.
Contributed* talk at the INFORMS Optimization Society Conference (IOS; Session: *Optimization of Complex Physics-Based Systems*), Houston, TX, 2024.

71. **CLAIRE: Scalable algorithms for diffeomorphic image registration**.
Contributed* talk at the SIAM Conference on Uncertainty Quantification (UQ24; Session: *Computational Tools for Large-Scale Inverse Problems and UQ*), Trieste, IT, 2024.

70. **Efficient algorithms for inverse problems governed by dynamical systems**.
Invited talk (host: K. B. Nakshatrala) at the Department of Civil and Environmental Engineering, University of Houston, TX, 2023.

69. **Fast algorithms for nonlinear optimal control of geodesic flows of diffeomorphisms**.
Contributed* talk at the U.S. National Congress on Computational Mechanics (USNCCM7; Session: *Recent Advances in Large-Scale Optimal Engineering Design*), Albuquerque, NM, 2023.

68. **Shape classification through the lens of geodesic flows of diffeomorphisms**.
Invited talk (hosts: D. Labate (UHouston), B. Tatiana (UBath), S. D. V. Rea (Centro de Investigación en Matemática)) at workshop entitled "Leveraging Model- and Data-Driven Methods in Medical Imaging" at Banff International Research Station for Mathematical Innovation and Discover, CA, 2023.

67. **Scalable algorithms for inverse problems governed by dynamical systems**.
Invited talk (host: C. M. Neuhauser) at DSI's webinar at the Hewlett Packard Enterprise Data Science Institute, University of Houston, Houston, TX, 2023.

66. **Deep neural networks for Bayesian inverse problems governed by nonlinear ODEs**.
Invited talk (hosts: A. S. Giovanni (UGenoa), A. Andrea (UMilan), B. Tatiana (UBath), R. Luca (UBologna), S. Matteo (UGenoa)) at workshop entitled "Learning for Inverse Problems" at Istituto Nazionale di Alta Matematica, Rome, IT, 2023.

65. **Fast algorithms for PDE-constrained optimization under uncertainty**.
Contributed* talk at SIAM Conference on Optimization (OP23; Session: *Challenges in Inverse Problems with Massive Data*), Seattle, US, 2023.

64. **Efficient numerical methods for optimal control problems governed by geodesic flows of diffeomorphisms**.
Invited colloquium talk (host: J. Rudi) at Department of Mathematics, Virginia Tech, Blacksburg, VA, US, 2023.

63. **Efficient numerical methods for optimal control problems governed by geodesic flows of diffeomorphisms**.
Invited talk (host: S. Foucart) at Center for Approximation and Mathematical Data Analytics, Texas A&M University, College Station, TX US, 2023.

62. **Fast algorithms for optimal control problems governed by geodesic flows of diffeomorphisms**.
Invited talk (host: S. Shontz) at Mathematical Methods and Interdisciplinary Computing Center (MMICC) at the University of Kansas, Lawrence, KS, US, 2023.

61. **Numerical methods for PDE-constrained optimization problems governed by hyperbolic equations**.
Invited colloquium talk (host: J. R. Romero) at the Department of Mathematical Sciences, University of Puerto Rico, Mayaguez Campus, US, 2023.

60. **Fast algorithms for inverse problems governed by transport equations**.
Contributed* talk at AMS Spring Southeastern Sectional Meeting Meeting (Session: *Recent Developments on Analysis and Computation for Inverse Problems for PDEs*) in Atlanta, GA, US, 2023.

59. **CLAIRE: Scalable multi-GPU algorithms for diffeomorphic image registration in 3D**.
Invited talk (host: G. Dogan) at Applied and Computational Mathematics Division (ACMD) Seminar Series at the National Institute of Standards and Technology (NIST), Boulder, CO, US, 2023.

58. **Deep learning for Bayesian inverse problems governed by nonlinear differential equations**.
Contributed talk at SIAM Conference on Computational Sciences and Engineering (CSE23; Session: *Uncertainty Quantification for Data-Intensive Inverse Problems and Learning*), Amsterdam, NL, 2023.

57. **Scalable algorithms for diffeomorphic image registration**.
Contributed talk at Dagstuhl Seminar *Inverse Biophysical Modeling and Machine Learning in Personalized Oncology*, Dagstuhl, Wadern, DE, 2023.

56. **CLAIRE: Scalable multi-GPU algorithms for diffeomorphic image registration**.
Contributed* talk at Joint Mathematics Meetings (JMM23; Session: *SIAM Minisymposium on Imaging and Inverse Problems*); Boston, MA, US, 2023.

55. **Fast algorithms for nonlinear optimal control of geodesic flows of diffeomorphisms**.
Invited talk (host: H. Antil) at CMAI Colloquium at George Mason University, Fairfax, VA, 2022.

54. **Fast algorithms for diffeomorphic image registration**.
Contributed* talk at SIAM Texas-Louisiana Sectional Meetings (SIAM TX-LA; Session: *Mathematics and Computation in Biomedicine*); Houston, TX, US, 2022.

53. **Randomized algorithms for efficient preconditioning and uncertainty quantification in inverse transport problems**.
Contributed* talk at SIAM Conference on Mathematics of Data Science (MDS22; Session: *Randomized Methods in Large-Scale Inference and Data Problems*); San Diego, CA, US, 2022.

52. **Fast algorithms for initial value control problems in diffeomorphic registration**.
Contributed talk at Copper Mountain Conference on Iterative Methods (CM22; Session: *Efficient Optimization Algorithms*); virtual conference, 2022.

51. **Fast algorithms for initial value control problems**.
Contributed* talk at SIAM Conference on Imaging Sciences (IS22; Session: *Partial Differential Equations and Control Problems*); virtual conference, 2022.

50. **CLAIRE: Scalable multi-GPU algorithms for diffeomorphic registration in 3D**.
Contributed talk at SIAM Conference on Optimization (OPT21; Session: *Large-Scale Optimization for Inverse Problems and Learning in Medical Imaging*); virtual conference, 2021.

49. **Uncertainty quantification in diffeomorphic image registration**.
Contributed* talk at SIAM Annual Meeting (AN21; Session: *Uncertainty Quantification Strategies for Data-Driven, Large-Scale Problems*); virtual conference, 2021.

48. **Uncertainty quantification for inverse transport problems**.
Contributed talk at SIAM Conference on Computational Science and Engineering (CSE21; Session: *Uncertainty Quantification for Data-Intensive Inverse Problems and Learning*); virtual conference, 2021.

47. **Fast algorithms for nonlinear optimal control of geodesic flows of diffeomorphisms**.
Invited talk (host: T. Bui-Thanh) at Oden Seminar at the Oden Institute for Computational Engineering and Sciences, University of Texas at Austin; virtual seminar; 2021.

46. **Statistical analysis of shapes and shape deformations in 3D**.
Contributed* talk at Joint Mathematics Meetings (JMM20; Session: *AMS Special Session on Geometry in the Mathematics of Data Science*); virtual conference, 2020.

45. **Classification of 3D shapes and shape deformations**.
Contributed talk at Annual Meeting of the SIAM Texas-Louisiana Section (SIAM TX-LA 20; Session: *Scientific Machine Learning*); virtual conference, 2020.

44. **Fast GPU-accelerated diffeomorphic image registration in 3D**.
Contributed* talk at SIAM Conference on Imaging Science (IS20; Session: *Fast Algorithms for Inverse Problems and their Applications*); virtual conference, 2020.

43. **Automatic classification of 3D shapes and shape deformations**.
Contributed* talk at SIAM Conference on Mathematics of Data Science (MDS20; Session: *Integration of Model-Based and Data-Based Methods with Medical Imaging*); virtual conference, 2020.

42. **Estimating oncogenic parameters via biophysical brain tumor growth modeling**.
Contributed* talk at Annual Meeting of the Society for Neuro-Oncology (Session: *Computational Neuro-Oncology*), Phoenix, AZ, US, 2019.

41. **Fast GPU-accelerated diffeomorphic image registration in 3D**.
Contributed talk at Annual Meeting of the SIAM Texas-Louisiana Section (SIAM TX-LA 19: *Recent Advances in Inverse Problems and Imaging*), Southern Methodist University Dallas, TX, US, 2019.

40. **Fast algorithms for nonlinear optimal control for diffeomorphic registration**.
Invited talk (host: R. Herzog & E. Trélat), Special Semester on Optimization organized by E. Sachs & K. Kunisch (Workshop: *New Trends in PDE-Constrained Optimization*), Johann Radon Institute for Computational and Applied Mathematics (RICAM), Linz, AT, 2019.

39. **Uncertainty quantification in non-linear optimal control problems for diffeomorphic registration**.
Contributed* talk (Session: *Uncertainty quantification strategies for physics applications*), AMS Fall Central Sectional Meeting, Madison, WI, US, 2019.

38. **Fast algorithms for nonlinear optimal control for diffeomorphic registration**.
Invited colloquium talk (host: C. Brune) at Department of Applied Mathematics, University of Twente, Enschede, NL, 2019.

37. **Fast diffeomorphic image registration in 3D**.
Contributed talk (Session: *Fast iterative methods for large-scale inverse problems in imaging*) at International Congress on Industrial and Applied Mathematics, Valencia, ES, 2019.

36. **Fast algorithms for initial value control problems in diffeomorphic image registration**.
Contributed* talk (Session: *Analysis and fast numerical methods for inverse problems and their applications*) at Applied Inverse Problems (AIP) Conference, Grenoble, FR, 2019.

35. **Diffeomorphic shape matching: Fast algorithms for non-linear optimal control problems**.
Invited talk (host: M. Mougeot) at Éléments de mathématique pour l'intelligence artificielle, École Normale Supérieure, Paris-Saclay, Cachan, FR, 2019.

34. **Optimal control of PDEs: Application to brain tumor modeling**.
Contributed* talk at AMS Southeastern Sectional Meeting (Session: *Validation and Verification Strategies in Multiphysics Problems*), University of Arkansas, Fayetteville, AR, US, 2018.

33. **Fast algorithms for inverse transport problems**.
Contributed talk (Session: *Inverse Problems*) at SIAM Annual Meeting, Portland, OR, US, 2018.

32. **CLAIRE: A parallel solver for constrained diffeomorphic image registration**.
Invited seminar talk (host: J. Kast) at Mint Medical GmbH, Heidelberg, DE, 2018.

31. **Parallel algorithms for hyperbolic PDE-constrained optimization problems**.
Contributed* talk (Session: *Krylov and regularization methods for large scale inverse problems*) at International Workshop on Parallel Matrix Algorithms and Applications (PMAA18), ETH Zuerich, Zuerich, CH, 2018.

30. **CLAIRE: A parallel solver for constrained large deformation diffeomorphic image registration**.
Invited talk (host: M. Schulte) at Department of Computer Science at University of Stuttgart, Stuttgart, DE, 2018.

29. **CLAIRE: A parallel solver for constrained large deformation diffeomorphic image registration**.
Contributed talk (Session: *Diffeomorphic image registration: Numerics, applications and theory*) at SIAM Conference on Imaging Sciences, Bologna, IT, 2018.

28. **Computational mathematics meets medicine: Formulations, numerics, and parallel computing**.
Invited talk (host: J. Nagy) at Numerical Analysis and Scientific Computing Seminar of Department of Mathematics & Computer Science, Emory University, Atlanta, GA, US, 2018.

27. **CLAIRE: A distributed-memory solver for constrained diffeomorphic image registration**.
Invited colloquium talk (host: J. Chan) at Computational and Applied Mathematics Department, Rice University, Houston, TX, US, 2018.

26. **Parallel algorithms for optimal control based diffeomorphic image registration**.
Contributed talk at Houston Imaging Sciences Symposium, Houston, TX, US, 2017.

25. **Preconditioners for the reduced space Hessian in hyperbolic optimal control problems**.
Contributed talk (Session: *Preconditioning methods in large-scale ill-posed inverse problems*) at International Conference on Preconditioning Techniques for Scientific and Industrial Applications, Vancouver, BC, CA, 2017.
24. **A distributed-memory Newton–Krylov solver for inverse transport problems**.
Contributed talk (Session: *Advances in computational methods for inverse problems*) at US National Congress on Computational Mechanics (USNCCM), Montreal, QC, CA, 2017.
23. **A distributed-memory Newton–Krylov solver for constrained diffeomorphic image registration**.
Contributed talk (Session: *Fast algorithms for numerical simulation and their applications in inverse problems*) at Applied Inverse Problems (AIP) Conference, Hangzhou, 2017.
22. **Parallel algorithms for PDE-constrained optimization problems with hyperbolic constraints**.
Contributed talk (Session: *Fast solvers for large-scale inverse problems in imaging*) at SIAM Conference on Computational Science and Engineering, Atlanta, GA, US, 2017.
21. **PDE-constrained optimization and high-performance computing meets medicine: Applications, formulations, and fast solvers**.
Invited talk (host: K. Josic) at Department of Mathematics, University of Houston, Houston, TX, US, 2016.
20. **A parallel solver for inverse transport problems**.
Invited talk (host: L. Ruthotto) at Emory University, Numerical Analysis and Scientific Computing Seminar, Department of Mathematics & Computer Science, Atlanta, GA, US, 2016.
19. **Fast solvers for constrained diffeomorphic image registration**.
Contributed talk (Session: *Efficient algorithms for large-scale inverse problems in medical imaging*) at SIAM Conference on Imaging Sciences, Albuquerque, NM, US, 2016.
18. **Fast algorithms for diffeomorphic image registration**.
Contributed talk at Workshop on Numerical Methods for Optimal Control and Inverse Problems, Technical University Munich, Munich, DE, 2016.
17. **Efficient evaluation of Hessian operators arising in large-scale inverse transport problems**.
Contributed talk (Session: *Uncertainty quantification and inversion of multiphysics and multiscale problems*) at SIAM Conference on Uncertainty Quantification, Lausanne, CH, 2016.
16. **Numerical simulation and optimization in medical imaging**.
Invited talk (hosts: A. Iske) at Department of Mathematics, University of Hamburg, Hamburg, DE, 2015.
15. **Efficient algorithms for optimal control based diffeomorphic image registration**.
Contributed talk at Workshop on Numerical Methods for Optimal Control and Inverse Problems, Technical University Munich, Munich, DE, 2015.
14. **Efficient algorithms for physically constrained diffeomorphic image registration**.
Contributed talk (Session: *Efficient algorithms for variational methods in imaging*) at SIAM Conference on Computational Science and Engineering, Salt Lake City, UT, US, 2015.
13. **Optimal control meets medical imaging sciences: Applications, formulations and fast solvers**.
Invited lecture (host: B. Menze) at Zentralinstitut für Medizintechnik (IMETUM), Technical University Munich, Munich, DE, 2015.
12. **Efficient algorithms for inverse transport problems**.
Invited talk (host: T. M. Buzug) at Graduate Seminar Mathematics in Image Computing and Simulation, University of Lübeck, Lübeck, DE, 2014.
11. **Computational tools in neuroimaging: From image computing to personalised modelling via PDE constrained optimisation**.
Invited talk at SIEMENS Molecular Imaging, Oxford, UK, 2012.

10. **Personalised image-based modelling of brain tumour progression via PDE constrained optimisation**.
    Contributed talk at Symposium Images and Networks of the Brain, University Medical Center Hamburg-Eppendorf, Hamburg, DE, 2012.
9. **From molecular signalling and cellular interactions to personalised modelling of brain cancer progression**.
    Contributed talk at the First Neuroimaging Workshop of the CBBM, University of Lübeck, Lübeck, DE, 2012.
8. **Image-based modeling of brain tumour progression: From individualization to priors for non-rigid image registration**.
    Contributed talk (Session: *Mathematical challenges in 4D imaging*) at SIAM Conference on Imaging Science, Philadelphia, PA, US, 2012.
7. **Image-based modelling of tumor growth**.
    Invited talk (host: J. Modersitzki) at Institute of Mathematics and Image Computing, University of Lübeck, Lübeck, DE, 2011.
6. **From image registration to bio-physical modelling: Analysis of brain tumour pathology**.
    Invited talk (host: W. Jessen and R. Nawrath) at 28. Workshop on Image Processing (*Arbeitskreis Bildverarbeitung*), University of Applied Sciences Kiel, Kiel, DE, 2010.
5. **A mathematical toolbox for the description of brain cancer dynamics**.
    Invited talk (host: M. Weiler) at German Cancer Research Center (*Deutsches Krebsforschungszentrum* (DKFZ)), Heidelberg, DE, 2010.
4. **Image registration with application to neuroradiology**.
    Invited talk (host: W. Jessen) at 23. Workshop on Image Processing (*Arbeitskreis Bildverarbeitung*), University of Lübeck, Lübeck, DE, 2008.
3. **Medical image registration with application to neuroradiology**.
    Contributed talk at Consortium for Medical Engineering (*Arbeitsgemeinschaft Medizintechnik in Schleswig-Holstein e.V.*), Lübeck, DE, 2008.
2. **Image registration with application to brain tumour imaging**.
    Invited talk (host: B. Fischer) at Institute of Mathematics, University of Lübeck, Lübeck, DE, 2007.
1. **Analysis of brain tumour growth via non-parametric non-rigid image registration**.
    Invited talk (host: J. Sabczynski) at Sector Medical Imaging Systems, Department Digital Imaging, Philips Research Europe, Hamburg, DE, 2007.

Teaching, Student Learning & Supervision Experience

Courses Taught

University of Houston

Course		Level	Students	Semester
MATH 6397	Comp. & Math. Methods in Data Science	GR	13	FA25
MATH 6367	Optimization Theory II	GR	24	SP25
MATH 2318	Linear Algebra	UG	60	SP24
MATH 6397	Comp. & Math. Methods in Data Science*	GR	26	SP24
MATH 6366	Optimization Theory I	GR	25	FA23
MATH 2318	Linear Algebra	UG	59	SP23
MATH 6397	Bayesian Inverse Problems & UQ*	GR	17	SP23
MATH 6366	Optimization Theory I	GR	20	FA22
MATH 3336	Discrete Mathematics	UG	80	SP22
MATH 3336	Discrete Mathematics	UG	100	FA21
MATH 6366	Optimization Theory I	GR	25	FA21
MATH 3336	Discrete Mathematics	UG	99	SP21
MATH 6397	Applied Inverse Problems*	GR	16	FA20
MATH 6366	Optimization Theory I	GR	11	FA20
MATH 2331	Linear Algebra	UG	75	SP20
MATH 6366	Optimization Theory I	GR	29	FA19
MATH 2331	Linear Algebra	UG	71	SP19
MATH 6366	Optimization Theory I	GR	20	FA18
MATH 2331	Linear Algebra	UG	36	SP17
MATH 2331	Linear Algebra	UG	81	FA17

University of Luebeck

Course		Level	Students	Semester
—	Image Processing	GR	20	SU12
—	Image Processing	GR	30	SU11
—	Image Processing*	GR	17	SU10
—	Imaging and Image Processing	GR	14	SU09
—	Imaging and Image Processing*	GR	15	SU08

∗: new course

Other Teaching Activities

2025 Principal Lecturer (8 lectures) at ChAMELEON Summer School, University of Houston, Houston, TX, USA

Student Mentoring

Postdoctoral Mentees

- **James L. Herring** (PhD: Applied Mathematics; Emory University), *Optimal Control & Inverse Problems*, 07/2018–12/2019.
  Current Employment: Research Scientist at Slingshot Aerospace, El Segundo, CA, USA

PhD Students

- **Christopher Chukwuemeka**, *Machine Learning*, since 08/2025.

- **Samundra Regmi** (co-advised w/N. Charon), *Machine Learning*, since 08/2024.
- **Pegah Amiri**, *Variational Bayesian Inference & Machine Learning*, since 08/2024.
- **Asikul Islam**, *Surrogate Modeling*, since 01/2024.
- **Jannatul Ferdous Chhoa** (co-advised w/Y. He), *CLAIRE-MG: Efficient Diffeomorphic Image Registration Using Multigrid-Preconditioned Newton–Krylov Methods* (https://hdl.handle.net/10657/20613), Department of Mathematics, University of Houston, 11/2021–07/2025
  Current Employment: Lecturer, Mathematics, University of St. Thomas, Houston, TX, US
- **Radmir Sultamuratov** (co-advised w/R. Azencott), *Distance-based classification and clustering of smooth surfaces* (https://hdl.handle.net/10657/18318), Department of Mathematics, University of Houston, 02/2021–11/2024
  Current Employment: Seismic Imaging Analyst at Viridien, Houston, TX, US.
- **German Villalobos**, *Scientific machine learning for Bayesian inverse problems governed by the FitzHugh–Nagumo model* (https://hdl.handle.net/10657/16107), Department of Mathematics, University of Houston, 08/2020–12/2023.
  Current Employment: Data Scientist at Layer Scaffolding, Houston, TX, USA.
- **Jae Youn Kim**, *Efficient numerical methods for initial value control problems for diffeomorphic image registration* (https://hdl.handle.net/10657/15999), Department of Mathematics, University of Houston, 08/2019–08/2023.
  Current Employment: Adjunct Lecturer, Department of Mathematics, University of Houston, TX, USA.
- **Saeed Sorena Sarmadi** (co-advised w/R. Azencott), *Stochastic neural networks for cell tracking in video recordings of bacterial colonies* (https://hdl.handle.net/10657/7991), Department of Mathematics, University of Houston, 03/2018–12/2020.
  Current Employment: Vice President of Artificial Intelligence, NthsDS, Houston, TX, USA.

BSc Theses

- **Philip Klein** (co-advised w/T. M. Buzug (ULuebeck)), *Rigide Registrierung von vektorwertigen Daten: Anwendungsgebiet MPI*, 2013.
- **Thomas Polzin** (co-advised w/T. M. Buzug (ULuebeck)), *Reorientation of diffusion tensors*, 2010.
- **Viktor Wottschel** (co-advised w/T. M. Buzug (ULuebeck)), *Diffusion-based filters for noise reduction in diffusion tensor imaging data*, 2010.

MSc Theses

- **Felix Huber** (co-advised w/M. Schulte (UStuttgart)), *Efficient algorithms for geodesic shooting in diffeomorphic image registration*, 2019.
- **Jenny Stritzel** (co-advised w/T. M. Buzug (ULuebeck)), *Modellindividualisierung als Optimierungsproblem mit Differenzialgleichungsnebenbedingung*, 2013.
- **Maik Stille** (co-advised w/W. R. Crum (KCL)), *Reconstruction of a 3-dimensional brain volume from fluorescent images and its co-registration with magnetic resonance imaging*, 2011.
- **Jan O. Jungmann** (co-advised w/T. M. Buzug (ULuebeck)), *In-silico modelling of tumour growth: Diffusion-based growth models*, 2009.
- **Stefan Becker** (co-advised w/T. M. Buzug (ULuebeck)), *In-silico modelling of tumour growth: Parametric deformation models*, 2009.
- **Leila Ghaderi** (co-advised w/T. M. Buzug (ULuebeck)), *Thin-plate spline based image registration*, 2007.

Undergraduate Research Projects

- **Dinh Binh Le** (SURF recipient), *Optimization & Deep Learning*, Summer 2025.
- **Anjalee Nair** (SURF recipient), *Optimization & Deep Learning*, Summer 2024.
- **Brayan A. Gutierrez** (SURF recipient), *Bayesian Inference*, Summer 2023.
- **Danial Khan** (SURF recipient), *Variational Bayesian Inference*, Summer 2022.
- **Gundeep Singh** (PURS recipient), *Inverse Problems*, Fall 2021.
- **Yaseen Syed** (PURS recipient), *Kernel Methods*, Spring 2021.
- **Syed Abidi** (SURF/PURS recipient), *Machine Learning*, Spring & Summer 2021.
- **Haley Rosso** (SURF recipient), *Machine Learning & Inverse Problems*, Summer 2020.
- **Gundeep Singh**, *Machine Learning*, Summer 2019.
- **Brenda Gonzalez** (2018 SURF recipient), *Machine Learning & Optimization*, Summer 2018.
- **Orion Lowy** (co-advised w/B. Bodmann (UHouston)), *Harmonic Analysis*, Summer 2018.
- **Naveen Himthani** (co-advised w/G. Biros (UTAustin)), *Numerical PDEs*, Summer 2015.
- **Krishan Mittal** (co-advised w/G. Biros (UTAustin)), *Numerical PDEs*, Summer 2014.
- **Ioanis Tsonas** (co-advised w/G. Biros (UTAustin)), *Image Computing*, Summer 2014.

PhD Committee Member

2025 **Raunak Sarbajna** (adviser: C. Eick (UHouston)): *Spatial computing frameworks for adaptive disaster response and visual place recognition*.

Michela Marini (adviser: D. Labate (UHouston)): *Supervised and unsupervised learning of single-cell morphological profiling*.

Navid Namakforoosh (adviser: W. Ott (UHouston)): *Theoretical and computational approaches to minimal mobile sensing*.

2024 **An Vu** (adviser: D. Labate (UHouston)): *Multi-dimensional orientable framelets with compact support*.

Sen Lin (adviser: R. Azencott (UHouston)): *Multi-quantile estimator: A robust estimation algorithm for generalized extreme value distribution parameters and its application in analyzing extreme values in intraday stock price movement*.

Ji Shi (adviser: D. Labate (UHouston)): *Optimal approximation of high-dimensional functions on smooth manifolds using deep ReLU neural networks*.

Heng Zhao (adviser: D. Labate (UHouston)): *Integration of model- and learning-based methods in image restoration*.

Phuong Tran (adviser: M. Nicol (UHouston)): *The study of extremes for functionals on certain hyperbolic dynamical systems: A theoretical and applied approach*.

2023 **Ivan Ezhov** (adviser: B. H. Menze (UZuerich)): *Learnable personalization of brain tumor growth models*.

Nickolas Fularczyk (adviser: D. Labate (UHouston)): *Deep learning for neuroscience imaging and convolutional framelets using a tensor representation*.

Yewen Huang (adviser: D. Labate (UHouston)): *Quantitative analysis of fluorescent images of glia cells using deep neural networks*.

Jennifer Ruhnow May (adviser: B. Bodmann (UHouston)): *Geometric conditions for the recovery of sparse signals on graphs from measurements generated with heat kernels*.

2022 **Seyyed Mahmood Ghasemi** (adviser: R. Azencott (UHouston)): *Gene transcription analysis at the cell population level*.

Alexander Zhiliakov (adviser: M. Olshanskii (UHouston)): *Trace finite element method for material surface flows*.

Yingxue Su (adviser: R. Azencott (UHouston)): *Rare events simulation in bacterial genetic evolution models*.

Naveen Himthani (adviser: G. Biros (UTAustin)): *High performance algorithms for medical image registration with applications in neuroradiology*.

2021 **Zahed Shahmoradi** (adviser: T. Lee (UHouston)): *Data-driven inverse linear programming: Integrating inverse optimization and machine learning*.

Jeric S. Alcala (adviser: I. Timofeyev (UHouston)): *Subgrid-scale parametrization of unresolved processes*.

Xiaoquian Chen (adviser: I. Timofeyev (UHouston)): *Effective data-driven models for chaotic and turbulent dynamics*.

Neil Jerome A. Egarguin (adviser: D. Onofrei (UHouston)): *Active control of exterior wavefields and applications*.

2020 **Wilfredo J. Molina** (adviser: D. Labate (UHouston)): *Improving the stability of the recovery of algebraic curves via Bernstein basis polynomials and neural networks.*

Basanta Raj Pahari (adviser: D. Labate (UHouston)): *Smooth projections and optimally sparse representation of cartoon-like cylindrical solids*.

Kazem Safaripoorfatide (adviser: D. Labate (UHouston)): *Receptive field convolutional neural networks and applications in image classification*.

Mohamadkazem Safaripoorfatide (adviser: D. Labate (UHouston)): *Receptive field convolutional neural networks and applications in image classification*.

Sabrine Hoteit Assi (adviser: D. Labate (UHouston)): *Geometric multiscale representations and applications to the analysis to retinal fundus images*.

2019 **Peng Zhang** (adviser: R. Azencott (UHouston) & J. He (UHouston)): *Diffeomorphic shape matching based on an operator splitting method*.

Kayla M. Bicol (adviser: A. Quaini (UHouston)): *Advances in algorithms for atmospheric sciences*.

2018 **Mikolaos Mitsakos** (M. Papadakis (UHouston)): *Virtual multimodal object detection and classification with deep CNNs*.

Professional Services & Synergistic Activities

Review Activity

Journals SIAM Journal on Scientific Computing [web] • SIAM Journal on Imaging Sciences [web] • SIAM Undergraduate Research Online [web] • Journal of Scientific Computing [web] • Journal of Computational Physics [web] • Applied Mathematics and Computation [web] • Applied Numerical Mathematics [web] • Journal of Inverse and Ill-Posed Problems [web] • AIMS Inverse Problems and Imaging [web] • AIMS Journal of Numerical Algebra, Control and Optimization [web] • IMA Journal of Applied Mathematics [web] • Transactions on Mathematical Software [web] • Parallel Processing Letters [web] • Optimization [web] • Optimization and Engineering [web] • Advances in Continuous and Discrete Models: Theory and Applications [web] • Computer Vision and Image Understanding [web] • Journal of Mathematical Imaging and Vision [web] • Computer Methods in Applied Mechanics & Engineering [web] • Mathematical & Computer Modelling [web] • Signal Processing [web] • IEEE Transactions on Medical Imaging [web] • NeuroImage [web] • Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences [web]

Grants Panelist for the National Science Foundation (NSF)

• Directorate for Mathematical and Physical Sciences (DMS; four panels).

• Directorate for Computer and Information Science and Engineering (CISE; one panel).

Dutch Research Council (NWO; one mail-in review)

Fellowships Panelist for the United States Department of Defense National Defense Science and Engineering Graduate (NDSEG) Fellowship (one panel)

Other Zentralblatt MATH since 2018 (https://zbmath.org; reviews).

Mathematical Reviews since 2020 (https://www.ams.org/mr-database).

Book review: "Introduction to the Tools of Scientific Computing" by Einar Smith, Springer Nature, 2020; featured review in SIAM Review, Vol. 64, No. 2, 2022.

Book review: "Inverse Optimal Control and Inverse Noncooperative Dynamic Game Theory. A Minimum-Principle Approach" by Timothy L. Molloy, Jairo Inga Charaja, Sören Hohmann & Tristan Perez, Springer, 2022 (https://zbmath.org/7477911).

Judge at multiple poster sessions (Workshop on Inverse Problems of the Association for Women in Mathematics (AWM) at SIAM Conference on Optimization (SIAM OP23) • AMS-PME undergraduate poster session at the Joint Mathematics Meetings (JMM 2023) • SIAM Conference on Mathematics of Data Science (MDS22) • MAA Student Poster Session of the Joint Mathematics Meetings (JMM 2021))

Program committee member (IEEE International Parallel & Distributed Processing Symposium (IPDPS'24) • MICCAI BrainLes Workshop 2022, a satellite event of the International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI) • MICCAI BrainLes Workshop 2021, a satellite event of the International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)).

Editorial Service

since 2023 Associate Editor of Advances in Continuous and Discrete Models (ACDM; [web])

since 2021 Associate Editor of the AIMS Journal of Numerical Algebra, Control and Optimization (NACO; [web])

Organization of Conferences, Workshops and Minisymposia

2026 Organizing Committee Co-Chair (w/R. Morrison (UColorado) & R. Willett (UChicago)) of **SIAM Conference on Mathematics of Data Science**, Salt Lake City, UT.

Co-organization (w/R. White (Sandia), T. Bui-Thanh (UTAustin) & L. Ramirez Ramirez (CIMAT)) of BIRS workshop Integrating Data- and Physics-Driven Methods for Decision Making under Uncertainty, Casa Matemática Oaxaca (CMO), Oaxaca, MX.

2025 Co-organization (w/L. Cappanera (UHouston), Y. He (UHouston), M. Wang (UHouston)) of CBMS Conference: Research at the Interface of Applied Mathematics and Machine Learning at the University of Houston, Houston, TX, US.

Co-organization (w/N. Charon (UHouston) and E. Hartman (UHouston) of mini-symposium **Recent advances in high-dimensional and geometric data analysis** at SIAM TX-LA Annual Conference, Austin, TX, US.

Organizer of ChAMELEON Summer School, University of Houston, Houston, TX, USA

Member of the SIAM **Coordinating Committee** for the Joint Mathematics Meetings (JMM 2026; Secretary).

Co-organization (w/A. Saibaba (NCState) of mini-symposium **Data- and Model-driven Approaches for Inverse Problems** at The Third Joint SIAM/CAIMS Annual Meetings, Montréal, Québec, CA.

Co-organization (w/J. Rudi (VTech) & I.-G. Farcas (VTech)) of mini-symposium **Fast Algorithms for Data- and Compute-Intensive Deterministic and Statistical Inverse Problems** at SIAM Conference on Computational Sciences and Engineering, Fort Worth, Texas, US.

2024 Co-organization (w/M. Chung (Emory)) of mini-symposium **Integration of Model and Data-Driven Methods for Large-Scale Inverse Problems** at SIAM Conference on Mathematics of Data Science, Atlanta, GA, US.

Co-organization (w/D. Onofrei (UHouston)) of mini-symposium **Data- and Model-Driven Approaches for Inverse Problems** at SIAM Texas-Louisiana Section Annual Meeting, Baylor University, Waco, TX, US.

Co-organization (w/A. Arnold (WPI)) of mini-symposium **Inverse Problems and Uncertainty Quantification in Biological and Medical Applications** at International Conference on Computational and Mathematical Biomedical Engineering, George Mason University, Arlington, VA, US.

Co-organization (w/J. Modersitzki (ULuebeck) & J. Lellmann (ULuebeck)) of mini-symposium **Model- and Data-Driven Approaches in Motion Analysis** at SIAM Conference on Imaging Sciences, Atlanta, GA, US.

Co-organization (w/E. Newman (Emory) & S. Gamble (SRNL)) of mini-symposium **Data-Driven Methods for Analysis and Understanding of Images** at SIAM Conference on Imaging Sciences, Atlanta, GA, US.

Co-organization (w/A. Arnold (Worcester Polytechnic Institute)) of mini-symposium **Uncertainty Quantification for Large-Scale Inverse Problems Governed by Dynamical Systems** at SIAM Conference on Uncertainty Quantification, Trieste, IT.

Organization of SIAM session **Mathematical Methods in Computer Vision and Image Analysis** at the Joint Mathematics Meeting, San Francisco, CA, US.

2023 Co-organization (w/A. Mamonov (UHouston) & D. Onofrei (UHouston)) of mini-symposium **Data- and Model-Driven Approaches for Inverse Problems** at SIAM TX-LA Annual Conference, Lafayette, LA, US.

Co-organization (w/G. Biros (UTAustin)) of mini-symposium **Recent Advances in Variational and Data-Driven Methods for Inverse Problems** at SIAM Conference on Optimization, Seattle, WA, US.

Co-organization (w/J. Rudi (VirginiaTech) & T. Bui-Thanh (UTAustin)) of mini-symposium **Uncertainty Quantification for Data-Intensive Inverse Problems and Learning** at SIAM Conference on Computational Sciences and Engineering, Amsterdam, NL.

Co-organization (with G. Biros (UTAustin), M. Schulte (UStuttgart), & B. Menze (ETH Zürich)) of Dagstuhl Seminar **Inverse Biophysical Modeling and Machine Learning in Personalized Oncology** (51-0120) at Schloss Dagstuhl, DE.

2022 Co-organization (w/A. Mamonov (UHouston) & D. Onofrei (UHouston)) of mini-symposium **Recent Advances in Large-Scale Inverse Problems: Numerics, Theory, and Applications** at SIAM TX-LA Annual Conference, Houston, TX, US.

Member of the organizing committee of the SIAM TX-LA Annual Conference at the University of Houston, Houston, TX (key responsibilities: maintenance of onlinepage; booklet and conference program).

Co-organization (w/D. Labate (UHouston)) of mini-symposium **Integration of Model- and Data-Driven Methods with Applications in Imaging** at SIAM Conference on Mathematics of Data Science, San Diego, CA, US.

Co-organization (w/T. Bui-Thanh (UTAustin)) of mini-symposium **Uncertainty Quantification for Data-Intensive Inverse Problems and Machine Learning** at World Congress on Computational Mechanics, Yokohama, JP.

Co-organization (w/G. Biros (UTAustin) & M. Schulte (UStuttgart)) of mini-symposium **Recent Advances on Integrating Optimization, Learning, and Modeling with Medical Imaging** at SIAM Conference on Imaging Sciences, Berlin, DE.

Co-organization (w/N. Charon (John Hopkins)) of mini-symposium **Shape Matching, Shape Analysis, and Morphometry: Theory, Numerics, and Applications** at SIAM Conference on Imaging Sciences, Berlin, DE.

Co-organization (w/G. Biros (UTAustin)) of mini-symposium **Large-Scale Optimization for Inverse Problems and Learning in Medical Imaging** at SIAM Conference on Optimization; virtual conference.

Co-organization (w/H. Goh (UTAustin) & T. Bui-Thanh (UTAustin)) of mini-symposium **Uncertainty Quantification for Data-Intensive Inverse Problems and Learning** at SIAM Conference on Computational Sciences and Engineering; virtual conference.

2020 Co-organization (w/S. Subramanian (UTAustin)) of mini-symposium **Recent Advances in Inverse Problems: Numerics, Theory, and Applications** at SIAM TX-LA Sectional Meeting; virtual conference.

Co-organization (w/D. Labate (UHouston)) of mini-symposium **Integration of Model-Based and Data-Based Methods with Medical Imaging** at SIAM Conference on Mathematics of Data Science; virtual conference.

2019 Co-organization (w/T. Bui-Thanh (UTAustin)) of mini-symposium **Recent Advances in Inverse Problems and Imaging** at SIAM TX-LA Sectional Meeting, Dallas, TX, US.

Co-organization (w/J. Herring (UHouston)) of mini-symposium **Fast Iterative Methods for Large-Scale Inverse Problems in Imaging** at International Congress on Industrial and Applied Mathematics, Valencia, ES.

Co-organization (w/J. Herring (UHouston)) of mini-symposium **Numerical Methods for Optimal Control Problems in Imaging** at Applied Inverse Problems Conference, Saint-Martin-d'Héres, FR.

Co-organization (w/J. Herring (UHouston) and G. Biros (UTAustin)) of mini-symposium **Fast Solvers for Inverse Problems with PDEs** at SIAM Conference on Computational Sciences and Engineering, Spokane, WA, US.

2018 Co-organization (w/G. Biros (UTAustin)) of mini-symposium **Diffeomorphic Image Registration: Numerics, Applications, and Theory** at SIAM Conference on Imaging Sciences, Bologna, IT.

2017 Co-organization (w/G. Biros (UTAustin)) of mini-symposium **Fast Solvers for Large-Scale Inverse Problems in Imaging** at SIAM Conference on Computational Sciences and Engineering, Atlanta, GA, US.

2016 Co-organization (w/G. Biros (UTAustin)) of mini-symposium **Efficient Algorithms for Large-Scale Inverse Problems in Medical Imaging** at SIAM Conference on Imaging Sciences, Albuquerque, NM, US

2015 Co-organization (w/G. Biros (UTAustin)) of mini-symposium **Efficient Algorithms for Variational Methods in Imaging** at SIAM Conference on Computational Sciences and Engineering, Salt Lake City, UT, US

University Service

- Member of **Executive Committee** (EC), Department of Mathematics, University of Houston, Houston, TX, US (academic years: 2024, 2025).
- Member of **Search Committee for Tenure-Track Positions**, Department of Mathematics, University of Houston, Houston, TX, US (five committees; 2019, 2020, 2021, 2022, 2023, 2024).
- Member of **Graduate Studies Committee** (GSC), Department of Mathematics, University of Houston, Houston, TX, US (academic years: 2020–2025).
- Member of **Undergraduate Studies Subcommittee** "Rebuilding the Math Community," Department of Mathematics, University of Houston, Houston, TX, US (academic year: 2023).
- Member of the **Organizing Committee of the Amundson Lecture Series**, Department of Mathematics, University of Houston, Houston, TX, US (academic year 2025).
- Co-organization (w/D. Labate) of **Data-Enabled Science Seminar** at the Department of Mathematics, University of Houston, Houston, TX, US (since 2020).
- Co-organization (w/D. Labate and M. Papadakis) of **Image Analysis Seminar** at the Department of Mathematics, University of Houston, Houston, TX, US (2018–2019).
- Judge for the Graduate Paper Presentations organized by the UH SIAM and UH AMS Student Chapters, Department of Mathematics, University Houston, Houston, TX, US (SP18, SP22, SP25).
- Faculty Adviser of Texas Theta Chapter of Pi Mu Epsilon (PME) (since 2018).
- Member of the **Colloquium Committee**, Department of Mathematics, University Houston, Houston, US (2018–2024).

- Member of **Search Committee for Faculty**, Division of Computer Science/Technology and Natural Sciences, University of Lübeck, Lübeck, DE (two committees 2011 & 2012).

Other Service

- Moderator of PME paper session VIII at Joint Mathematics Meeting 2024 (Friday, January 5 2023)

Academic and Professional Memberships

since 2014 Society of Industrial and Applied Mathematics (SIAM)

since 2015 Association for Computing Machinery (ACM)

since 2018 American Mathematical Society (AMS)

since 2018 Pi Mu Epsilon (PME)

Outreach

SP23 Participation in **Math and Science Family Night** at the Durham Elementary School, Houston, TX.

SP20 Talk **Making Waves with Mathematics** at Mega-Mathletes event organized by the Mu Alpha Theta National Honor Society at the Stephen F. Austin High School, Houston, TX

References

References available on request.

Houston TX, 10/22/2025

(*Andreas Mang*)